

ABDUL HANAN

Associate Software Engineer

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SUMMARY

Results-driven Software Engineer experienced in building and deploying production-grade backend systems, RESTful APIs, and cloud-native applications. Proficient in Python, FastAPI, PostgreSQL, Docker, and AWS with a strong foundation in system design, automation workflows, and full SDLC. Proven ability to deliver scalable, maintainable solutions from API architecture to containerized cloud deployments.

EDUCATION

FAST National University of Computer & Emerging Sciences

B.S. in Artificial Intelligence | August 2022 – May 2026

EXPERIENCE

AI/ML Engineer & Python Instructor — Contract

Cognara Training | November 2025 – April 2026 | FastAPI, LangChain, Docker, AWS, PostgreSQL, pgvector

- Engineered two production systems—APD Listener Tool and Tradie Migration Platform—covering full backend architecture, REST API design, JWT/RBAC auth, and AWS cloud deployment.
- Built LangChain RAG pipeline with pgvector for AI-powered document search; deployed via Docker on AWS EC2 behind Nginx with multithreaded queue-based processing.
- Delivered Python curriculum (fundamentals through backend dev, APIs, automation, and AI/ML) to international students.

AI Engineer Intern

One Tech and AI | July 2025 – September 2025 | RAG, LLMs, Docker, FastAPI

- Architected and deployed RAG-based chatbots and AI agents end-to-end, owning retrieval pipeline design, LLM orchestration, and production API services.
- Containerized microservices with Docker and FastAPI; contributed to applied Quantum AI research over a 3-month engagement.

SKILLS & TOOLS

Languages & Frameworks: Python, C++, SQL, Java, FastAPI, REST APIs, PostgreSQL, Neo4j

AI / ML: LangChain, RAG, Hugging Face Transformers, FAISS, pgvector, PyTorch, TensorFlow

Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker, IAM), Docker, Nginx, GitHub Actions CI/CD, MLflow, Airflow

SELECTED PROJECTS

Tradie Migration Platform

FastAPI, PostgreSQL, LangChain, OpenAI, pgvector, Docker, AWS | Mar – Apr 2026

- Role-based SaaS migration platform with 6-role RBAC/JWT auth, LangChain+pgvector RAG for AI document retrieval, and Gmail SMTP/OTP automated workflows.
- Deployed containerized backend on AWS EC2/Nginx; integrated OpenAI APIs for intelligent visa case and candidate matching.

APD Listener Tool

React, FastAPI, Docker, AWS EC2/S3, Nginx, Vercel | Nov 2025 – Feb 2026

- End-to-end audio-to-instructions platform for an Australian client; multithreaded queue backend on EC2/Nginx, media via S3, React frontend on Vercel.

Doctor Path AI — Clinical Decision Support

pgvector, Neo4j, FAISS, Docker, FastAPI, MedGemma | Aug 2025 – Feb 2026

- Clinical AI system on MIMIC-III with hybrid retrieval: pgvector embeddings + Neo4j knowledge graphs + FAISS similarity ranking for treatment recommendations.

Hybrid E-Commerce Recommendation System

FAISS, Annoy, CLIP, PyTorch, MiniLM | Mar – May 2025

- Multimodal recommendation engine fusing MiniLM text + CLIP image embeddings; two-stage Annoy→FAISS retrieval for high-accuracy product search at scale.

HONORS & ACTIVITIES

- Top 50 Nationally — ICPC 2026 | Head of Job Fair 2026, Co-Head 2025, Coordinator 2024 — FAST NU
- Teaching Assistant: Generative AI, Deep Learning, DB Systems, Assembly Language — FAST NU CFD | 350+ LeetCode problems
- Python Instructor — Taught international students Python (beginner to advanced): OOP, automation, REST APIs, web scraping, backend development, and AI/ML concepts | July 2025 – January 2026
- Level 2 Fiverr Freelancer — AI Automation & Backend Development